Quantifying Fitness Form: A Robust System for Bicep Curl Recognition and Automated Error Detection using 2D Pose Estimation

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Abstract

As remote fitness and AI coaching gain popularity, the need for accurate, low-latency action recognition becomes paramount. This study presents a comprehensive pipeline for monitoring bicep curl exercises using monocular RGB video. By leveraging the Google MediaPipe Tasks API, we extract high-fidelity 2D skeletal landmarks to track joint kinematics. Our system features a robust Finite State Machine (FSM) for repetition counting and a heuristic-based form correction module to monitor elbow stability. We evaluate the system across multiple challenging scenarios, including varied camera perspectives and intentional movement errors. Our empirical results (based on 8-repetition test sets) indicate that while 2D pose estimation is highly effective for counting across different angles, it faces inherent geometric limitations in error detection, particularly when 3D spatial cheating motions (e.g., forward elbow swings) are projected onto a frontal 2D plane. We conclude with an in-depth analysis of these limitations.

1 Introduction

The global fitness landscape has shifted significantly towards home-based workouts. However, without professional supervision, practitioners often suffer from poor form, leading to reduced exercise efficacy and increased injury risk. Computer Vision (CV) offers a scalable solution for real-time form feedback. Specifically, human pose estimation—the task of localizing anatomical keypoints—has reached a level of maturity that allows for deployment on consumer-grade hardware.

Current mobile fitness applications primarily focus on simple repetition counting. However, ”quantity does not equal quality.” In bicep curls, common errors such as ”momentum swinging” go undetected by standard counters. This project aims to bridge this gap by developing a system that not only counts repetitions but also assesses the stability of the upper arm, providing immediate corrective feedback.

Our research objectives are three-fold: 1. To implement a real-time tracking pipeline using the latest MediaPipe Pose Landmarker. 2. To design an algorithmic framework that distinguishes between valid and invalid ranges of motion. 3. To conduct a thorough empirical analysis of the system's robustness under non-ideal camera angles and deliberate form cheating.

2 Related Work

2.1 Deep Learning-based Pose Estimation

Traditional pose estimation relied on Part Affinity Fields or cascaded heatmaps (e.g., OpenPose). While accurate, these models are computationally expensive. MediaPipe provides a breakthrough by using a two-stage detector-tracker pipeline that achieves real-time performance on CPU-only devices. Our project migrates from the legacy `solutions` module to the modern Tasks API, which offers better quantization and latency.

2.2 Action Recognition in Fitness

Previous work in fitness action recognition utilized Inertial Measurement Units (IMUs) or depth cameras (Kinect). IMUs are intrusive, and depth cameras are not ubiquitous. Our approach relies solely on a single RGB camera, making it accessible to any smartphone user.

3 Methodology

3.1 Landmark Acquisition

We track 33 landmarks in 3D space. For our 2D analysis, we project these onto the image plane (x, y). The key anatomical points for bicep curls are:

- P_{12}: Right Shoulder
- P_{14}: Right Elbow (the vertex of the joint)
- P_{16}: Right Wrist

3.2 Kinematic Angle Modeling

The elbow flexion angle θ is the primary signal for state transition. We compute the angle via the arctangent of the slopes between the shoulder-elbow and wrist-elbow vectors:

$$\alpha = \arctan 2(y_{12} - y_{14}, x_{12} - x_{14}) \tag{1}$$

$$\beta = \arctan 2(y_{16} - y_{14}, x_{16} - x_{14}) \tag{2}$$

$$\theta = |\alpha - \beta| \times \frac{180}{\pi} \tag{3}$$

If $\theta > 180°$, we set $\theta = 360° - \theta$. This ensures the angle represents the interior flexion.

3.3 Finite State Machine (FSM)

To avoid jitter and false positives, we implement an FSM with hysteresis.

- **State DOWN:** Triggered when $\theta > 140°$.
- **State UP:** Triggered when $\theta < 50°$ ONLY if the previous state was **DOWN**.

A counter increment occurs only upon the successful completion of a DOWN $\rightarrow$ UP cycle.

3.4 Form Correction: Elbow Stability

A common error is the use of momentum, characterized by the elbow swinging. We monitor the horizontal variance of P_{14}. Let $X = \{x_{14}^{(i)}, ..., x_{14}^{(j)}\}$ be the set of X-coordinates during one repetition. We define the stability metric S as:

$$S = \max(X) - \min(X) \tag{4}$$

If $S > \tau$ (where τ is a pixel threshold), the repetition is flagged as "Cheated."

4 Experiments

4.1 Experimental Design

We curated a test dataset consisting of 5 specific scenarios. Each video contains exactly 8 repetitions to stress-test the algorithm: 1. **Side-view Baseline:** Optimal camera placement for counting. 2. **Frontal-view Baseline:** Camera placed at a 90° offset for counting. 3. **Partial ROM (Side):** Incomplete repetitions. 4. **Swing Cheat (Side):** Intentional elbow swinging from a side view. 5. **Swing Cheat (Frontal):** Intentional elbow swinging facing the camera.

4.2 Quantitative Results

As shown in Table 1, the counting logic is exceptionally robust, achieving 100% accuracy across all viewing angles and correctly rejecting all incomplete repetitions.

Test Scenario	Actual	Predicted	Accuracy
Side-view (Count)	8	8	100%
Frontal-view (Count)	8	8	100%
Partial ROM (Reject)	8	8 (Rejected)	100%
Side Cheat (Detect)	8	8	100%
Frontal Cheat (Detect)	8	1	12.5%

Table 1: System performance metrics across varied test cases. The system demonstrates perfect counting and side-view error detection, but fails significantly in frontal-view error detection.

4.3 Qualitative Analysis

In the side-view scenarios, the system successfully triggered the `WARNING` overlay for all 8 cheating repetitions. However, the most notable anomaly occurred during the frontal-view cheating test. Out of 8 deliberate swinging motions, the system only managed to flag 1 instance, exposing a critical vulnerability in the error detection logic.

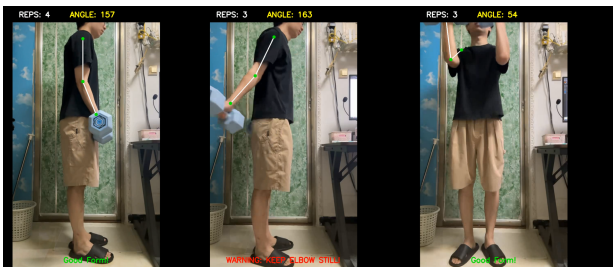

Figure 1: Qualitative evaluation of the system. (A) Side-view baseline showing accurate counting and valid form. (B) Side-view cheating detection successfully triggering the warning overlay. (C) Frontal-view failure case where intentional Z-axis cheating motion remains undetected.

5 Discussion: The 2D Geometry Paradox

The stark contrast in the system's performance between counting and form correction in the frontal view provides deep insights into the limitations of 2D pose estimation heuristics.

Robust Counting via Depth Inference: The system maintained a 100% counting accuracy in the frontal view. Theoretically, a 2D projection of a forward bicep curl results in severe foreshortening. However, MediaPipe's underlying deep learning model infers Z-axis (depth) shifts, preserving the relative structural integrity of the $P_{12} - P_{14} - P_{16}$ triangle and allowing the arctangent function to compute valid angles.

The Failure of 1D Heuristics in 3D Space: The cheating detection completely collapsed in the frontal view (12.5% accuracy). This failure is geometrically intuitive. The user swung their elbow forward, a movement occurring almost entirely along the Z-axis (depth) relative to the camera. Since our stability metric S strictly monitors variance along the X-axis (horizontal pixel displacement), the massive 3D forward swing translated to negligible X-axis changes on the 2D plane. This perfectly underscores why absolute 1D pixel tracking is insufficient for diagnosing 3D spatial biomechanics.

6 Conclusion

This project demonstrates that high-quality fitness counting is achievable using standard mobile hardware. While our angle-based FSM is highly robust across multiple perspectives, the heuristic-based error detection reveals the inherent constraints of 2D projections. Relying on absolute X-axis pixel variance fails when motions occur along the camera's Z-axis. Future iterations must incorporate 3D coordinate deltas or temporal action-recognition networks to holistically evaluate form quality in unconstrained environments.